Jiarui Hai

+1-224-866-2166 | jhai2@jhu.edu | haidog-yaqub.github.io

LinkedIn | Github | Google Scholar

Research Interest

My research focuses on AI for audio and speech signal processing, with a particular emphasis on generative models and the integration of natural language prompting. Specifically, I am interested in text-to-audio, text-to-voice generation, text-to-speech synthesis, text-guided source separation, and audio understanding tasks such as general audio question answering and text-guided sound event detection.
In addition, my background and passion for music production inspire a complementary interest in music technology, including music generation, transcription, source separation and restoration, and singing voice synthesis.

Education

- **Johns Hopkins University** — *Baltimore, MD, USA*
  Ph.D Student in Electrical and Computer Engineering — Aug 2022 - Expected 2026
  Advisor: Prof. Mounya Elhilali
- **Tsinghua University** — *Beijing, China*
  M.E. in Civil Engineering — Aug 2020 - Jun 2022
  Member in Big Data Program
- **Tsinghua University** — *Beijing, China*
  B.E. in Civil Engineering — Aug 2016 - Jun 2020
  B.S. in Business Analytics

Experience

- **Adobe CAVA | Speech AI Lab** — *San Francisco, CA, USA*
  Research Intern | Supervisors: Ke Chen, Rithesh Kumar, Zeyu Jin — May 2025 - Present
  Research Topic: Speech Synthesis
- **Johns Hopkins University | LCAP** — *Baltimore, MD, USA*
  Research Assistant | Supervisor: Mounya Elhilali — Aug 2022 - Present
  Research Topic: Audio Generation and Understanding
- **Tencent AI Lab | Speech Group** — *Bellevue, WA, USA*
  Research Intern | Supervisors: Yong Xu, Hao Zhang, Dong Yu — May 2024 - Aug 2024
  Research Topic: Audio Generation
- **Kuaishou | AI Platform** — *Beijing, China*
  Algorithm Engineer Intern | Supervisors: Rui Lu, Zhiyao Duan — Jun 2020 - Jan 2021
  Research Topic: Singing Voice Synthesis

Selected Publications

FULL LIST AVAILABLE ON GOOGLE SCHOLAR

[12] **Jiarui Hai**, Helin Wang, Weizhe Guo, Mounya Elhilali. “FlexSED: Towards Open-Vocabulary Sound Event Detection". WASPAA 2025 (**Oral**).

[11] **Jiarui Hai**, Mounya Elhilali. “SynSonic: Augmenting Sound Event Detection through Text-to-Audio Diffusion ControlNet and Effective Sample Filtering". WASPAA 2025.

[10] Yongyi Zang, **Jiarui Hai**, Wanying Ge, Helin Wang, Zheqi Dai, Yuki Mitsufuji, Qiuqiang Kong, Mark Plumbley. “The Inaugural Music Source Restoration Challenge". ICASSP 2026 Grand Challenge.

[9] Helin Wang*, **Jiarui Hai***, Dading Chong, Karan Thakkar, Tiantian Feng, Dongchao Yang, Junhyeok Lee, Laureano Moro Velazquez, Jesus Villalba, Zengyi Qin, Shrikanth Narayanan, Mounya Elhiali, Najim Dehak. “CapSpeech: Enabling Downstream Applications in Style-Captioned Text-to-Speech". Preprint.

[8] Helin Wang, **Jiarui Hai**, Dongchao Yang, Chen Chen, Kai Li, Junyi Peng, Thomas Thebaud, Laureano Moro-Velazquez, Jesus Villalba, Najim Dehak. “SoloSpeech: Enhancing Intelligibility and Quality in Target Speech Extraction through a Cascaded Generative Pipeline". Preprint.

[7] **Jiarui Hai**, Yong Xu, Hao Zhang, Chenxing Li, Helin Wang, Mounya Elhilali, Dong Yu. “EzAudio: Enhancing Text-to-Audio Generation with Efficient Diffusion Transformer". IntersSpeech 2025 (**Oral**).

[6] Helin Wang*, **Jiarui Hai***, Yen-Ju Lu, Karan Thakkar, Mounya Elhilali, Najim Dehak: “SoloAudio: Target Sound Extraction with Language-oriented Audio Diffusion Transformer". ICASSP 2025.

[5] **Jiarui Hai***, Karan Thakkar*, Helin Wang, Zengyi Qin, Mounya Elhilali. “DreamVoice: Text-Guided Voice Conversion". Interspeech 2024.

[4] **Jiarui Hai***, Helin Wang*, Dongchao Yang, Karan Thakkar, Najim Dehak, Mounya Elhilali. “DPM-TSE: A Diffusion Probabilistic Model for Target Sound Extraction". ICASSP 2024.

[3] **Jiarui Hai***, Yu-jeh Liu*, Mounya Elhilali. "Boosting Modality Representation with Pre-trained Models and Multi-task Training for Multimodal Sentiment Analysis". ASRU 2023.

[2] **Jiarui Hai**, Mounya Elhilali. "Diff-Pitcher: Diffusion-based Singing Voice Pitch Correction". WASPAA 2023 (**Oral**).

[1] Rui Lu, Baigong Zheng, **Jiarui Hai**, Fei Tao, Zhiyao Duan, Ji Liu. "Progressive Teacher-Student Training Framework for Music Tagging". ICASSP 2022.

RESEARCH ACTIVITIES

- **Peer Reviewer**
  - Conference: ICASSP, InterSpeech, WASPAA, ASRU, SLT
  - Journal: IJCV
  - Workshop: DeLTa @ ICLR 2025, Audio Imagination @ NeurIPS 2024
- **Open-source Contribution**
  - **OpenSound** User-friendly live demos for diverse audio and speech models: https://huggingface.co/OpenSound
  - **MeanFlow-PyTorch** A few-step flow-matching model training pipeline: https://github.com/haidog-yaqub/MeanFlow
- **Datasets, Benchmarks, and Challenges**
  - **Music Source Restoration Challenge** A benchmark for evaluating music source restoration systems: https://msrchallenge.com/
  - **CapSpeech** A general speech captioning dataset: https://wanghelin1997.github.io/CapSpeech-demo/
  - **DreamVoice** A voice timbre dataset for speaker identity modeling and synthesis: https://haidog-yaqub.github.io/dreamvoice_demo/

MUSIC ACTIVITIES

- **Professional Recognition**
  - Featured Producer on a rap single showcased in HIPHOP BANK (TV show, China) | Aug 2021
  - Invited Lecture on Music Production, Modern Sky (Chinas leading music company) | Jun 2021
  - Top New Producer Award, BeatsHome Hip-hop Production Contest | Apr 2021
- **Creative & Media Activities**
  - Independent Artist & Beatmaker: produce singles and beat tapes
  - Content Creator: share tutorials on music production and mixing
- **Leadership and Co-founder**
  - President, Rap & HipHop Association, Tsinghua University | 2020–2021
  - Co-founder, InnerSup Records (Music Producer Team)
  - Co-founder, Music Production Team, Rap & HipHop Association, Tsinghua University